

# RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: \_\_\_\_\_

PRN / Roll No: \_\_\_\_\_

Division: \_\_\_\_\_

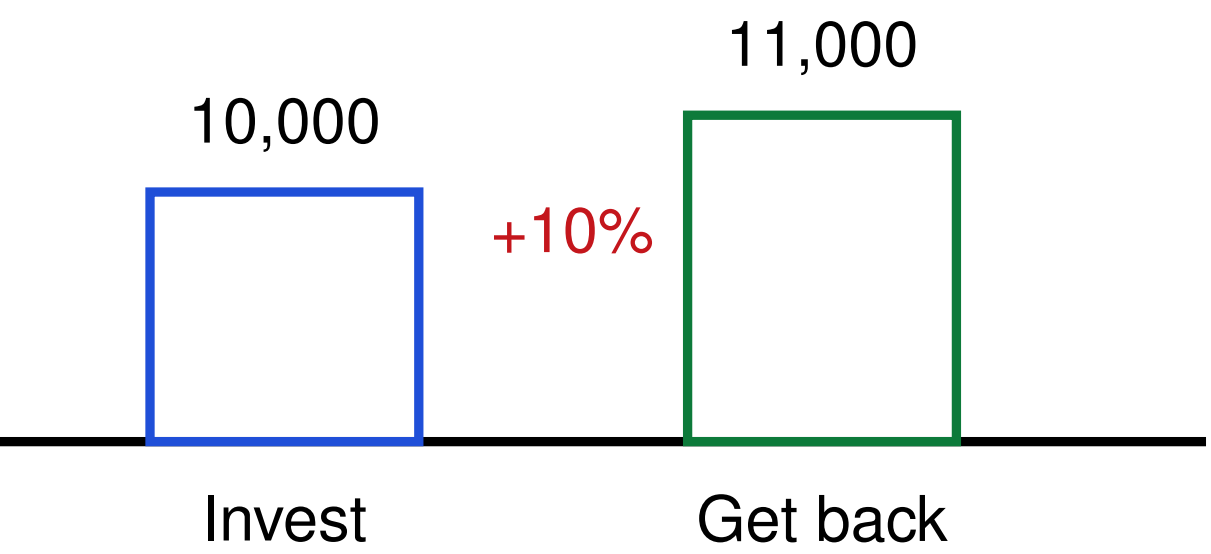
## UNIT V: BASICS OF INVESTMENT

### 1. Investment Return

The gain (income + price rise) on an investment, as a % of the money invested.

$$\text{Return} = \frac{\text{Gain}}{\text{Invested}} \times 100$$

Eg: Invest Rs 10,000, get back Rs 11,000.  
Return =  $1,000 \div 10,000 = 10\%$



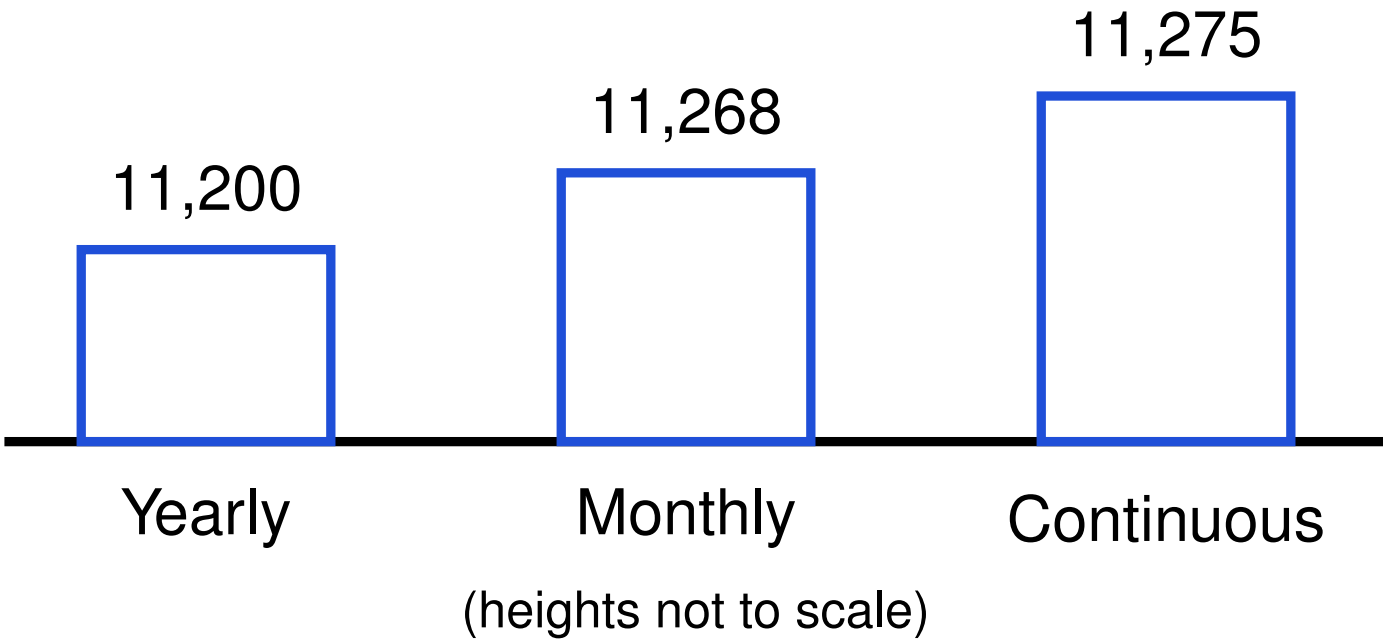
### 2. Compounding Frequency

At the same yearly rate, adding interest more often gives a little more money.

$$A = P \left( 1 + \frac{r}{m} \right)^{mt}$$

Continuous:  $A = Pe^{rt}$

Eg: Rs 10,000 at 12% for 1 year

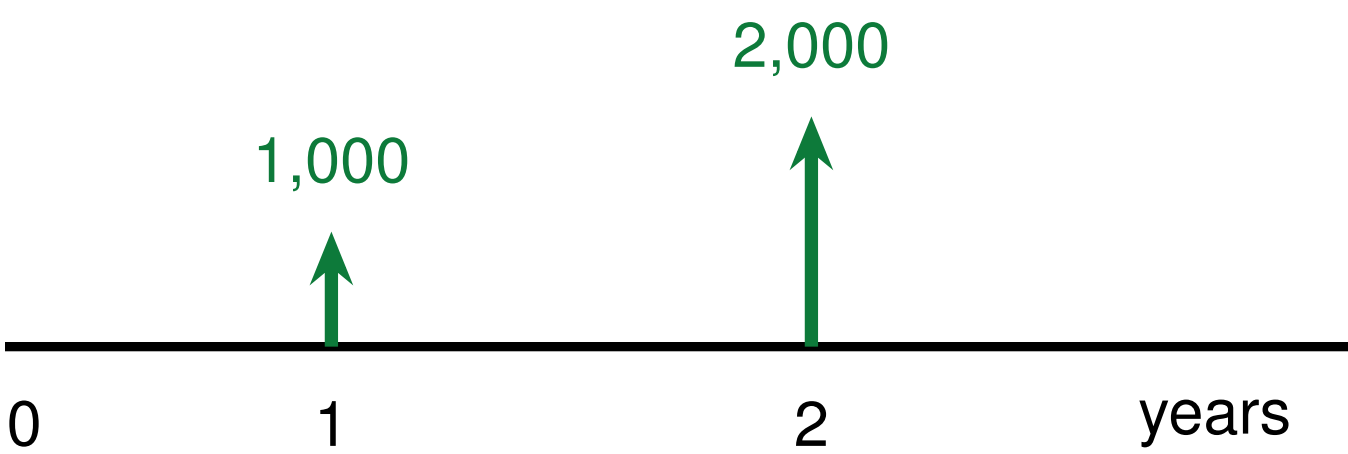


### 3. Uneven Cash Flows

Money received differs each year. Discount each to today and add.

$$PV = \sum \frac{CF_t}{(1+i)^t}$$

Eg: Rs 1,000 (year 1) and Rs 2,000 (year 2),  $i = 10\%$   
 $PV = 909 + 1,653 = \text{Rs } 2,562$

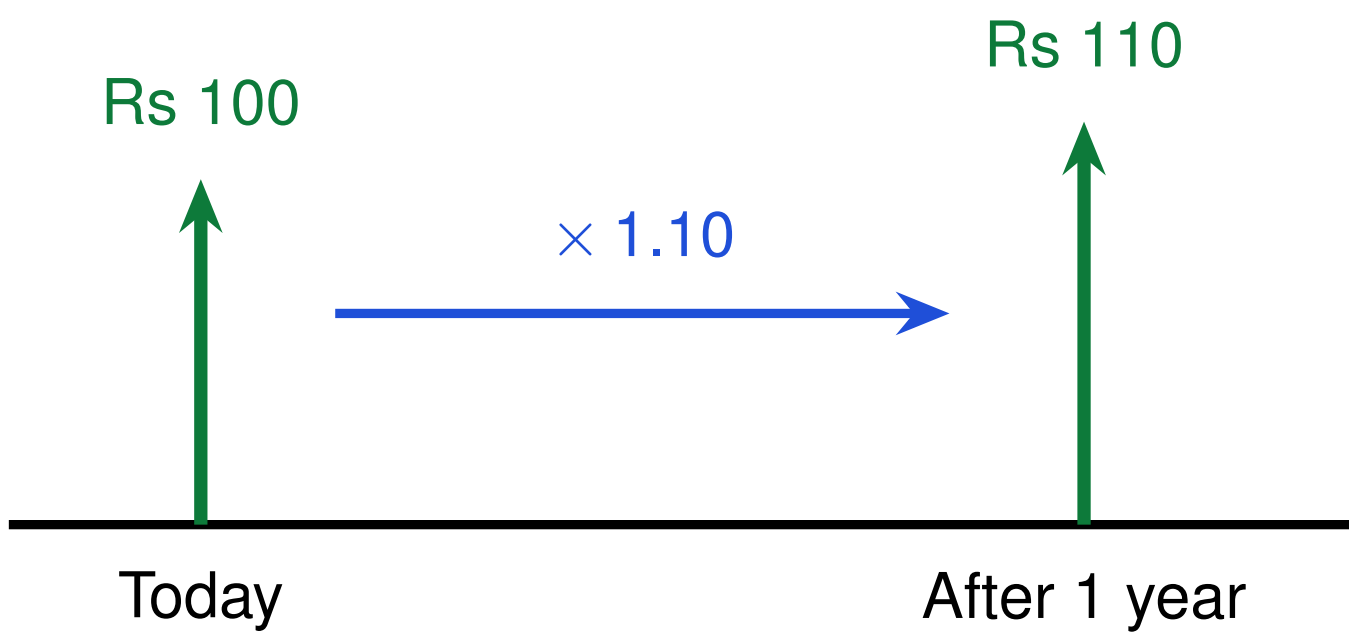


### 4. Economic Equivalence

At a given interest rate, different sums at different times can have the same value.

$$F = P(1+i)^n \quad P = \frac{F}{(1+i)^n}$$

Eg: at 10%, Rs 100 today = Rs 110 after 1 year



## UNIT VI: RISK AND UNCERTAINTY

### 1. Decision under Risk

Chances are known. Choose the best average (EMV).

$$\text{EMV} = \sum (\text{chance} \times \text{payoff})$$

Eg: 50%: Rs 100, 50%: Rs 40  $\Rightarrow \text{EMV} = \text{Rs } 70$



### 2. Decision under Uncertainty

Chances are unknown. Use a rule:

- **Maximax**: best of the best  $\rightarrow$  **Stock**
- **Maximin**: best of the worst  $\rightarrow$  **Bond**
- **Min-max regret**: least regret  $\rightarrow$  **Bond**

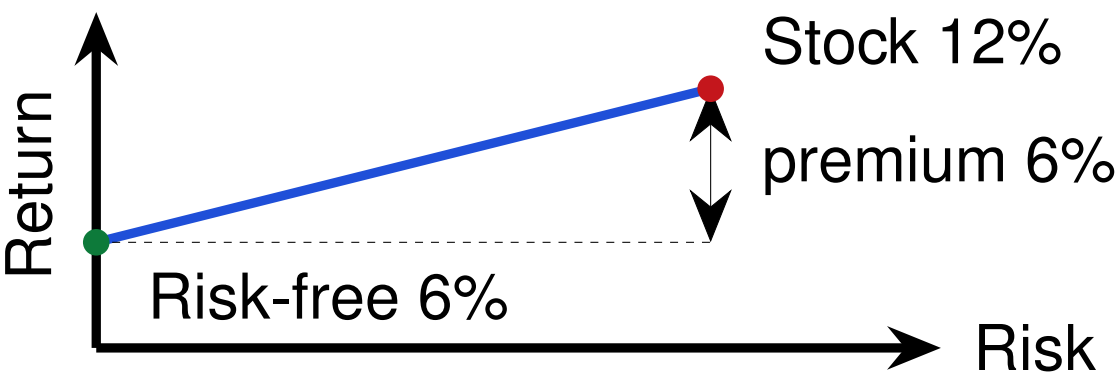
| Payoff | Boom | Slump |
|--------|------|-------|
| Stock  | 80   | -20   |
| Bond   | 40   | 25    |

### 3. Risk Premium

The extra return a risky asset must offer over a risk-free one.

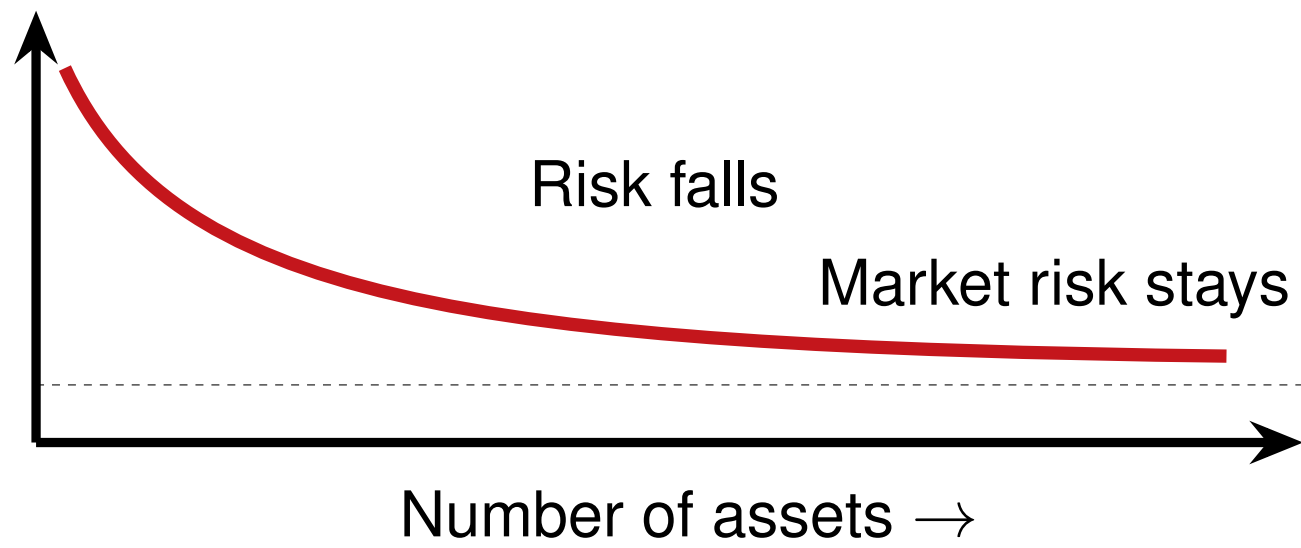
$$\text{Premium} = \text{Expected} - \text{Risk-free rate}$$

Eg:  $12\% - 6\% = 6\%$



### 4. Portfolio Diversification

Spread money over assets that do not move together to lower risk. Market risk cannot be removed.



### 5. Life Insurance

You pay a premium. The insurer pays the sum assured if you die within the term.

$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg (1 yr):  $\text{Rs } 1,00,000 \times 0.01 = \text{Rs } 1,000$



### 6. Endowment

Pays the sum assured if you **die in the term** OR **survive to the end**.

$$\text{Endowment} = \text{Term} + \text{Pure endowment}$$

Pure endowment = paid only if you survive.



Investors demand a higher expected return for taking higher risk